HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented validation strategies for laboratory IT systems within a **GMP-regulated environment**, ensuring alignment with **data integrity** and regulatory requirements.
- Collaborated with **Quality Control (QC)**, IT, and project teams to execute validation protocols and maintain system compliance, reducing critical findings by **15%** in the last audit cycle.
- Developed and maintained comprehensive **validation documentation**, including IQ, OQ, and PQ protocols, for laboratory instruments and associated software.
- Automated report generation for system performance monitoring using **Python scripting**, reducing manual effort by approximately **8 hours per week** for the operations team.
- Managed and resolved production incidents for enterprise applications, adhering to strict SLAs and conducting root cause analysis using **Splunk** and **Dynatrace**.
- Contributed to the development of **Java Spring Boot** REST APIs for cross-service data exchange, implementing robust error handling and request validation.
- Containerized laboratory IT applications using **Docker** and deployed them to an **AWS EC2** environment, improving deployment consistency and reducing setup time by **40%**.

SKILLS

Validation – CSV, IQ, OQ, PQ, GMP, Data Integrity, Regulatory Compliance

Laboratory Systems – QC Systems, LIMS, Chromatography Data Systems (CDS)

Programming & Scripting – Python, Java, Spring Boot, SQL

Cloud & DevOps – AWS (EC2, S3, Lambda), Docker, Jenkins, Git

Monitoring & Observability – Splunk, Dynatrace, CloudWatch

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

GMP Laboratory System Validation Framework

- Developed a standardized framework for the Installation Qualification (IQ), Operational Qualification (OQ), and Performance Qualification (PQ) of laboratory IT systems, ensuring consistent validation across diverse equipment.
- Created automated Python scripts to parse and validate system logs against predefined integrity requirements, flagging potential deviations for immediate review and reducing manual log analysis time by **30%**.
- Documented detailed validation protocols and executed test cases for a new chromatography data system, ensuring compliance with GMP standards and successful handover to the QC operations team.

Automated Production Monitoring Dashboard

- Engineered a Python-based data pipeline to ingest operational metrics from **Splunk** and **Dynatrace**, consolidating them for enhanced system monitoring.
- Built and deployed a web-based dashboard using **Flask** and **PostgreSQL** to visualize key performance indicators and alert thresholds for the production support team, aiding in faster incident detection.
- Integrated the dashboard with a notification system, reducing average alert response time by approximately **22%** through more immediate and targeted issue identification.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)